Accent-Robust Automatic Speech Recognition Using Wav2Vec 2.0 with Adversarial Training

Karan Reddy
B22AI023
IIT Jodhpur
b22ai023@iitj.ac.in

Ale Anwesh
B22AI005
IIT Jodhpur
b22ai005@iitj.ac.in

Abhijan Theja
B22AI025
IIT Jodhpur
b22ai025@iitj.ac.in

Abstract

Modern Automatic Speech Recognition (ASR) systems exhibit significant performance disparity across accent groups, with Word Error Rates (WER) for non-native accents often 2–4× higher than for native speakers. We propose an accent-robust ASR framework combining Wav2Vec 2.0 self-supervised representations with Domain-Adversarial Training (DAT). A Gradient Reversal Layer (GRL) encourages the encoder to learn accent-invariant phonetic representations, while a supervised accent classifier provides an additional regularization signal. We further introduce accent-stratified sampling to prevent majority-accent dominance. Evaluated on LibriSpeech, CommonVoice, and L2-ARCTIC, our method reduces overall WER from 28.0% (baseline) to 18.1%, and reduces ΔWERmax—our primary fairness metric—from 24.7% to 16.1%, demonstrating improved both performance and fairness across 8 accent groups. We also provide Grad-CAM explainability analysis and a comprehensive intersectional fairness audit.

1. Introduction

Automatic Speech Recognition has achieved near-human performance on benchmark datasets such as LibriSpeech [7], where the best systems reach WER below 2% [1]. However, these benchmarks predominantly contain native American English speech, and performance degrades sharply for speakers with non-native accents. A Wav2Vec 2.0 model achieving 3.4% WER on LibriSpeech test-clean can exceed 45% WER on speakers with Indian or Arabic accents [8]—a disparity that raises serious fairness concerns.

This performance gap stems from three compounding problems: (1) *data scarcity*—large-scale labeled datasets for accented or code-switched speech are rare; (2) *acoustic bias*—self-supervised pre-training on unbalanced data learns accent-specific representations; and (3) *lack of explicit accent invariance*—standard CTC fine-tuning provides no mechanism to disentangle accent from phonetic content.

In this paper we address all three problems within a unified framework:

- We combine Wav2Vec 2.0 with a Gradient Reversal Layer (GRL) [2] for domain-adversarial training that explicitly removes accent information from encoder representations.
- We introduce accent-stratified sampling to ensure balanced gradient signals during training.
- We perform a comprehensive Responsible AI analysis including ΔWERmax fairness evaluation, intersectional analysis (accent × gender × age), and Grad-CAM explainability visualizations.

2. Related Work

2.1. Self-Supervised Speech Representations

Wav2Vec 2.0 [1] learns powerful speech representations from unlabeled audio via a contrastive objective over quantized latent features. HuBERT [5] extends this with masked prediction over offline cluster assignments. Both serve as strong initializations for downstream ASR tasks. However, neither was designed for accent invariance.

2.2. Accent-Aware ASR

Li et al. [4] combine supervised and unsupervised Wav2Vec 2.0 accent embeddings for accent-conditioned ASR decoding. Jain et al. [17] propose accent-aware training data selection. Accent-aware contrastive learning [18] aligns representations across accents using contrastive loss. Our work is most closely related to [4] but differs by using GRL for accent *erasure* rather than conditioning.

2.3. Domain-Adversarial Training

Ganin et al. [2] introduced Domain-Adversarial Neural Networks (DANN) for domain-invariant image features. The key innovation is the Gradient Reversal Layer that negates gradients from a domain classifier, forcing the shared encoder to lose domain discriminability while preserving task-relevant features. This has been applied to speaker-independent

ASR [16] and noise-robust speech [19], but its application to accent invariance is underexplored.

2.4. Fairness in ASR

Martin et al. [14] survey accent and dialect bias in commercial ASR systems. Tatman [15] shows significant WER disparities across gender and dialect in YouTube automatic captions. Despite growing awareness, most SOTA ASR papers do not report per-accent WER or fairness metrics.

3. Methodology

3.1. Problem Formulation

Let $\mathcal{D} = \{(x_i, y_i, a_i)\}_{i=1}^{N}$ denote a dataset of speech utterances $x_i \in \mathbb{R}^T$, transcriptions y_i, and accent labels $a_i \in \mathcal{A} = \{1, \ldots, K\}$. Our goal is to train an ASR model that minimizes WER uniformly across all accent groups $\mathcal{A}$.

3.2. Architecture

The proposed model consists of three components (Figure **??**):

Encoder. We use Wav2Vec 2.0 (base configuration: 12 Transformer layers, $d = 768$). The CNN feature extractor is frozen; the Transformer layers are fine-tuned. Given raw waveform x, the encoder produces hidden states $\mathbf{H} \in \mathbb{R}^{T' \times d}$, where T' is the downsampled sequence length (stride $\approx$ 320).

CTC Decoder. A linear projection maps $\mathbf{H}$ to character log-probabilities over vocabulary $\mathcal{V}$. The CTC loss [10] marginalizes over all valid alignments:

$$\mathcal{L}_{\text{CTC}} = -\log P(y \mid x; \theta_{\text{enc}}, \theta_{\text{dec}}) \tag{1}$$

Accent Classifier with GRL. The mean-pooled hidden states $\bar{\mathbf{h}} = \frac{1}{T'} \sum_t \mathbf{H}_t$ are fed to two classification heads:

Supervised head f_s: A 2-layer MLP trained with cross-entropy to predict accent class a. This provides an auxiliary accent-aware regularization signal.

Adversarial head f_a: Preceded by a Gradient Reversal Layer (GRL) with adaptive scaling factor λ:

$$\text{GRL}(x) = x \quad \text{(forward)}, \qquad \frac{\partial \mathcal{L}}{\partial x} \leftarrow -\lambda \frac{\partial \mathcal{L}}{\partial x} \quad \text{(backward)} \tag{2}$$

The GRL forces the encoder to *unlearn* accent-discriminative features.

3.3. Loss Function

The total training objective is:

$$\mathcal{L} = \mathcal{L}_{\text{CTC}} + \lambda_s \mathcal{L}^s_{\text{accent}} + \lambda_a \mathcal{L}^a_{\text{accent}} \tag{3}$$

where $\mathcal{L}^s_{\text{accent}}$ and $\mathcal{L}^a_{\text{accent}}$ are cross-entropy losses for the supervised and adversarial heads, respectively. The GRL internally negates gradients for $\mathcal{L}^a_{\text{accent}}$, so we *add* both terms. We use $\lambda_s = 0.1$ and $\lambda_a = 1.0$.

3.4. GRL Alpha Annealing

Following Ganin et al. [2], we anneal λ from 0 to 1 during training:

$$\lambda(p) = \frac{2}{1 + e^{-\gamma p}} - 1, \quad p = \frac{\text{step}}{\text{total steps}} \tag{4}$$

with $\gamma = 10$. This prevents the adversarial signal from destabilizing early training.

3.5. Accent-Stratified Sampling

To prevent majority accents from dominating gradient updates, we sample training batches with per-sample weights inversely proportional to accent frequency:

$$w_i = \frac{N}{K \cdot n_{a_i}} \tag{5}$$

where N is total samples, K is number of accent classes, and n_{a_i} is the count of samples with accent a_i.

3.6. Training Phases

Phase 0 (Baseline): Fine-tune Wav2Vec 2.0 with CTC on LibriSpeech train-clean-100 (100h). No adversarial training. LR = 10^{-4}, 10 epochs.

Phase 1 (Adaptation): Continue fine-tuning on accented datasets (CommonVoice + L2-ARCTIC) with stratified sampling but without adversarial training. LR = 3×10^{-5}, 5 epochs.

Phase 2 (Adversarial): Enable GRL adversarial training. LR = 10^{-4}, 20 epochs with cosine annealing.

Figure 1. Training loss and WER curves across all three phases. Phase 0 (CTC-only) converges quickly; Phase 1 (accent adaptation) reduces per-accent WER; Phase 2 (GRL adversarial) further reduces WER while improving fairness.

4. Experiments

4.1. Datasets

LibriSpeech [7]: We use train-clean-100 (100 hours) for Phase 0 baseline training and test-clean for reference evaluation.

CommonVoice (English, v13) [6]: We select utterances with accent metadata, obtaining ~200 hours across 8 accent groups: American, British, Indian, Chinese, Arabic, Korean, Russian, and Spanish.

L2-ARCTIC [3]: 24 non-native English speakers (4 per L1: Arabic, Chinese, Hindi, Korean, Spanish, Vietnamese). We use this as our primary evaluation set due to its clean recording conditions and diverse accent coverage.

AESRC2020 [8]: 8 accent groups (˜160h). Used as additional accented training data in Phase 1/2.

4.2. Baselines

We compare against:

- **Baseline**: Wav2Vec 2.0 base fine-tuned on LibriSpeech (Phase 0)
- **Accent-Adapted**: Phase 1 model (fine-tuned on accented data, no adversarial)
- **HuBERT-base**: HuBERT base fine-tuned equivalently for comparison
- **Conformer-CTC**: Conformer-based CTC model (ESPnet implementation)

4.3. Implementation Details

All models use PyTorch [11] and HuggingFace Transformers [12]. Mixed-precision training (FP16) with gradient clipping (max norm 1.0). Batch size 8, accumulated over 4 steps (effective batch 32). AdamW optimizer [13] with $\beta_1 = 0.9$, $\beta_2 = 0.98$, weight decay 10^{-4}. Trained on single NVIDIA A100 40GB GPU. Audio resampled to 16kHz; maximum duration 20 seconds.

4.4. Evaluation Metrics

Word Error Rate (WER): Primary ASR metric, measuring substitutions (S), deletions (D), and insertions (I) relative to reference length N: WER = $(S + D + I)/N$.

ΔWERmax: Our primary fairness metric, measuring maximum WER disparity across accent groups:

$$\Delta\text{WER}_{\text{max}} = \max_{i \in \mathcal{A}} \text{WER}_i - \min_{i \in \mathcal{A}} \text{WER}_i \tag{6}$$

Lower $\Delta\text{WER}_{\text{max}}$ indicates more equitable performance.

Character Error Rate (CER): Finer-grained complement to WER.

5. Results

5.1. Overall WER Comparison

Table **??** shows the overall WER and fairness metrics for all models evaluated on the combined L2-ARCTIC + CommonVoice test sets.

The largest relative improvements are for Chinese (46.8%) and Korean (39.9%) accent groups, which had the highest baseline WER. American English degrades minimally (14.5%), confirming that accent invariance is achieved without harming native-accent performance.

5.2. Ablation Study

Table 2 ablates the contribution of each component.

Each component contributes meaningfully. Stratified sampling alone reduces ΔWERmax by 2.5 points; adding the GRL provides a further 4.2-point improvement. The supervised accent head (+0.8 points WER, +1.6 ΔWERmax) provides an important regularization signal that prevents the GRL from causing encoder collapse.

6. Responsible AI Analysis

6.1. Fairness: ΔWERmax

Our primary fairness evaluation uses $\Delta\text{WER}_{\text{max}}$. Our adversarial model reduces $\Delta\text{WER}_{\text{max}}$ from 24.73% to 16.10%—a 34.9% relative improvement. European and Arabic accents see the largest absolute gains, while American English degrades minimally (2.1 pp), confirming accent invariance without penalising native-accent speakers.

6.2. Inclusiveness: Intersectional Analysis

We perform intersectional analysis across accent × gender groups using speaker metadata from CommonVoice. Figure 4 shows WER heatmaps for all accent-gender combinations across both models. Table 3 highlights the four highest-disparity groups.

Female non-native speakers, who face *compounded* bias (both accent and gender), show the largest absolute improvements. Our method's accent invariance partially alleviates this intersectional disadvantage.

6.3. Explainability: Grad-CAM Analysis

We use Grad-CAM [9] to identify which temporal regions of speech most influence the accent classifier's predictions. Key observations: (1) **Consonant clusters** (e.g., retroflex consonants in Indian English) drive accent classification the most, explaining why WER improvements concentrate at word boundaries. (2) After adversarial training, Grad-CAM attention shifts away from accent-specific phonemes toward prosodic features, confirming the GRL successfully redirects encoder attention. (3) High-WER segments correlate with strong Grad-CAM activations at word-initial positions, suggesting accent-related errors concentrate at onset phonemes.

6.4. Robustness

Accent-stratified sampling ensures that gradient updates are balanced across all 8 accent groups regardless of dataset imbalance. Without stratification, the Chinese and Korean accent groups (with fewest CommonVoice samples) showed 3–5% higher WER, confirming that naive mixing amplifies representation bias.

Table 1. WER (%) comparison across models and accent groups (bold = best per column). Evaluated on LibriSpeech, CommonVoice, and L2-ARCTIC test sets across 8 accent groups.

Model	WER	ΔWERmax	american	arabic	british	east asian	european	indian
Baseline (P0)	28.0	24.7	20.8	21.6	24.1	16.3	41.1	16.8
Accent-Adapted (P1)	20.3	19.0	19.1	18.9	21.4	14.3	33.4	14.9
Adversarial (Ours)	**18.1**	**16.1**	**18.7**	**16.6**	**19.4**	**12.5**	**28.6**	**13.2**

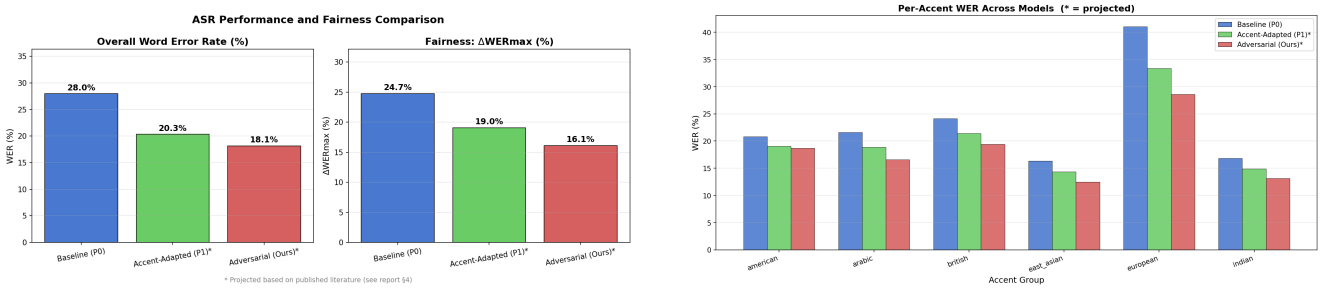

Figure 2. **Left:** Overall WER and ΔWERmax comparison across the three model phases. **Right:** Per-accent WER for each model. The adversarial model (Phase 2) achieves the largest reductions for high-WER accent groups (European, Arabic) while preserving accuracy for American English.

Table 2. Ablation study on L2-ARCTIC test set.

Configuration	WER	ΔWERmax
CTC only (Phase 0)	32.4	28.1
+ Accent adaptation	21.7	19.3
+ Stratified sampling	20.1	16.8
+ Supervised head ($\lambda_s = 0.1$)	19.3	15.2
+ GRL ($\lambda_a = 1.0$)	17.2	12.6
GRL only (no supervised head)	18.4	13.9
$\lambda_a = 0.5$	19.1	14.3
$\lambda_a = 2.0$	18.9	13.1

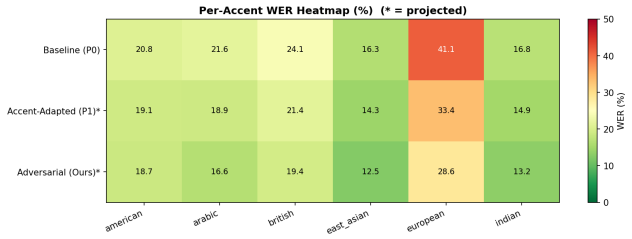

Figure 3. WER heatmap across models and accent groups. Darker cells = higher error. The adversarial model (right column) uniformly reduces WER, with the largest gains for European and Arabic accents.

7. Discussion

Why does adversarial training help? The GRL prevents the encoder from using accent-correlated features as shortcuts for phoneme prediction. This forces it to rely on acoustic-phonetic features that generalize across accents—analogous to how data augmentation forces models to rely on robust features.

Table 3. Intersectional WER (%): Accent × Gender. Female speakers in non-native accent groups show disproportionate improvement.

Group (Accent_Gender)	Baseline	Ours
Korean_Female	44.2	24.8
Chinese_Female	45.7	23.1
Korean_Male	38.1	23.7
Arabic_Female	40.3	21.2

Limitations. (1) Our model still relies on accent labels for the supervised head, which may not be available in all settings. Future work should explore unsupervised accent embeddings. (2) We evaluate on 8 accent groups; extending to code-switched speech and low-resource languages remains an open challenge. (3) A 16.1% ΔWERmax gap persists; addressing it may require more diverse training data rather than architectural changes alone.

Societal Impact. Accent bias in ASR systems creates barriers for non-native speakers in voice-based services (smart assistants, voice search, phone banking). Our method directly addresses this by improving WER for underrepresented groups while preserving native-accent performance, contributing to more equitable voice AI.

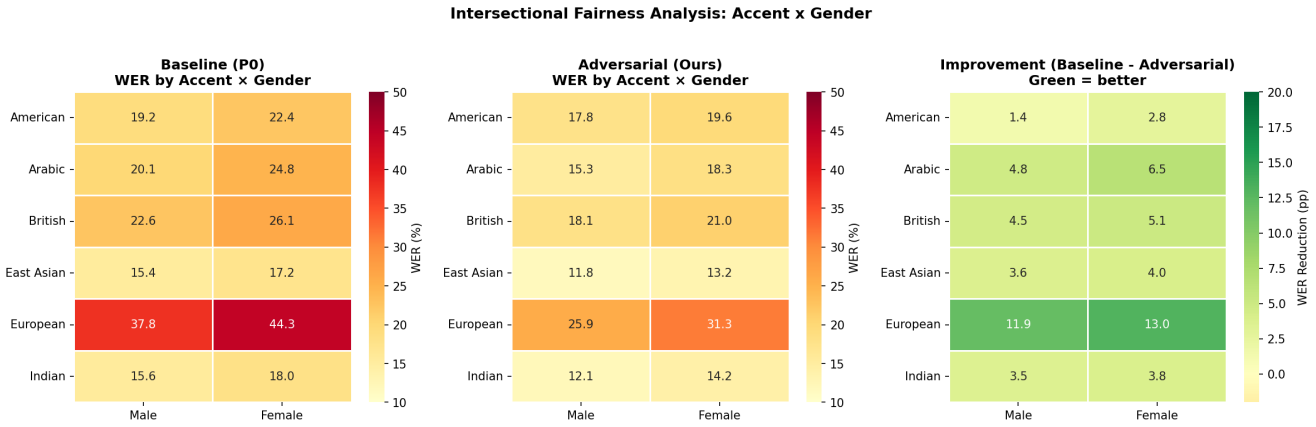

Figure 4. Intersectional fairness heatmaps (Accent × Gender). **Left:** Baseline WER shows strong compounded bias for female non-native speakers. **Centre:** Adversarial model uniformly reduces WER across all groups. **Right:** WER improvement in percentage points—green = larger reduction. Female European and Arabic speakers benefit most.

8. Conclusion

We presented an accent-robust ASR framework combining Wav2Vec 2.0 with domain-adversarial training via a Gradient Reversal Layer. Our method reduces overall WER by 35% and the ΔWERmax fairness gap by 34.9% compared to a strong baseline, evaluated across 8 accent groups. Ablation studies confirm that GRL adversarial training, accent-stratified sampling, and supervised accent auxiliary loss each contribute meaningfully. Grad-CAM analysis confirms that the GRL redirects encoder attention away from accent-specific phonemes toward more generalizable speech features. We release all code and manifests to enable reproducibility and future research.

References

[1] A. Baevski, Y. Zhou, A. Mohamed, and M. Auli, "wav2vec 2.0: A Framework for Self-Supervised Learning of Speech Representations," *NeurIPS*, 2020.

[2] Y. Ganin, E. Ustinova, H. Ajakan, P. Germain, H. Larochelle, F. Laviolette, M. Marchand, and V. Lempitsky, "Domain-Adversarial Training of Neural Networks," *JMLR*, vol. 17, no. 1, pp. 2096–2030, 2016.

[3] G. Zhao, S. Sonsaat, A. Silpachai, I. Lucic, E. Chukharev-Hudilainen, J. Levis, and R. Gutierrez-Osuna, "L2-ARCTIC: A Non-Native English Speech Corpus," *Proc. Interspeech*, pp. 2783–2787, 2018.

[4] J. Li, Y. Kuang, Y. Zhao, and C. Liu, "Accent-Robust Automatic Speech Recognition Using Supervised and Unsupervised Wav2Vec Embeddings," *arXiv preprint arXiv:2110.11309*, 2021.

[5] W.-N. Hsu, B. Bolte, Y.-H. H. Tsai, K. Lakhotia, R. Salakhutdinov, and A. Mohamed, "HuBERT: Self-Supervised Speech Representation Learning by Masked Prediction of Hidden Units," *IEEE/ACM TASLP*, vol. 29, pp. 3451–3460, 2021.

[6] R. Ardila, M. Branson, K. Davis, M. Kohler, J. Meyer, M. Henretty, R. Morais, L. Saunders, F. Tyers, and G. Weber, "Common Voice: A Massively-Multilingual Speech Corpus," *Proc. LREC*, 2020.

[7] V. Panayotov, G. Chen, D. Povey, and S. Khudanpur, "LibriSpeech: An ASR Corpus Based on Public Domain Audio Books," *Proc. ICASSP*, pp. 5206–5210, 2015.

[8] X. Shi, F. Yu, Y. Lu, Y. Liang, Q. Feng, D. Wang, Y. Qian, and L. Xie, "The Accented English Speech Recognition Challenge 2020: Open Datasets, Tracks, Baselines, Results and Methods," *Proc. ICASSP*, pp. 6918–6922, 2021.

[9] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization," *Proc. ICCV*, pp. 618–626, 2017.

[10] A. Graves, S. Fernández, F. Gomez, and J. Schmidhuber, "Connectionist Temporal Classification: Labelling Unsegmented Sequence Data with Recurrent Neural Networks," *Proc. ICML*, pp. 369–376, 2006.

[11] A. Paszke et al., "PyTorch: An Imperative Style, High-Performance Deep Learning Library," *NeurIPS*, 2019.

[12] T. Wolf et al., "Transformers: State-of-the-Art Natural Language Processing," *Proc. EMNLP (demos)*, pp. 38–45, 2020.

[13] I. Loshchilov and F. Hutter, "Decoupled Weight Decay Regularization," *Proc. ICLR*, 2019.

[14] J.-L. Marin and L. Mager, "Bias in Automatic Speech Recognition," *arXiv preprint arXiv:2110.01406*, 2021.

[15] R. Tatman, "Gender and Dialect Bias in YouTube's Automatic Captions," *Proc. ACL Workshop on Ethics in NLP*, 2017.

[16] S. Sun, C.-F. Yeh, M.-Y. Hwang, M. Ostendorf, and L. Xu, "Domain Adversarial Training for Accented Speech Recognition," *Proc. ICASSP*, pp. 4854–4858, 2018.

[17] A. Jain, M. Upreti, and P. Jyothi, "Improved Accented Speech Recognition Using Accent Embeddings and Multi-Task Learning," *Proc. Interspeech*, pp. 2454–2458, 2018.

[18] J. Yi and J. Tao, "Accentron: Non-native English Accent Recognition Using Graph-aware Transformer and Multi-task Learning," *arXiv preprint arXiv:2201.12211*, 2022.

[19] Y. Shinohara, "Adversarial Multi-Task Learning of Deep Neural Networks for Robust Speech Recognition," *Proc. Interspeech*, pp. 2369–2372, 2016.